RTKLIB-B2b User Manual

A Real-Time Decoding Module for BDS-3 PPP-B2b in RTKLIB

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1.Introduction

The BeiDou Global Navigation System (BDS-3) provides Precise Point Positioning (PPP) service via B2b signals, known as PPP-B2b, in China and its surrounding regions. Currently, the services provides precise satellite orbit and clock offsets for BDS and GPS constellations. Extensive researches have been conducted in evaluating the accuracies of individual products, as well as the performance in user positioning (Sun et al. 2023; Yangyin Xu, Yang, and Li 2021; Chen et al. 2025). These results suggest that the PPP-B2b service can provide accurate and robust services for users and is suitable for precise applications such as marine positioning, time transfer, and troposphere monitoring (Ying Xu, Zhao, and Wang 2024; Wang et al. 2024; Zhou et al. 2024; Geng et al. 2022).

The signal structure of PPP-B2b is identical to that of the B2b signal, both are broadcast on the I component. Upon capturing these messages, receivers internally demodulate the signal and subsequently output the decoded information as binary code to the user via the serial port or other communication interfaces (CNSO,2020). While the BDS B2b Interface Control Document (ICD) defines a standard format for this binary code, receiver manufacturers often modify it for proprietary reasons. This software is designed to decode these manufacturer-specific PPP-B2b binary correction messages and conduct PPP. It currently supports receivers from Sino (ComNavTech, SinoGNSS, also referred to as Comnav) and Unicore (Unicore Communications). Additionally, current RTCM standard has not yet incorporated BDS CNAV1 message formats, this software, for user convenience, also includes decoders for GPS LNAV and BDS CNAV1 ephemerides provided in the proprietary binary formats of SINO and UNICORE receivers.

This document is structured into the following sections to familiarize users with the functionalities of the RTKLIB-B2b software:

（1） RTKLIB-B2b Structure, Compilation, and Installation

（2） RTKLIB-B2b Decoder

（3） RTKLIB-B2b Positioning

（4） Software Maintenance and Data SharingAppendix

（5） References and Appendix

2. RTKLIB-B2b Structure, Compilation, and Installation

(1) RTKLIB-B2b Software Structure

RTKLIB-B2b is a software tool developed on the basis of RTKLIB to decode PPP-B2b messages and performing PPP. It currently supports decoding proprietary PPP-B2b message formats, as well as GPS LNAV and BDS CNAV1 ephemerides, from receivers manufactured by SINO (e.g., K803W) and UNICORE (e.g., UM980).

Fig 1 illustrates the schematic diagram of the RTKLIB-B2b decoder, distinguishing between the file-based input path for post-processing and the data stream-based input path for real-time processing, along with their respective core decoding steps.

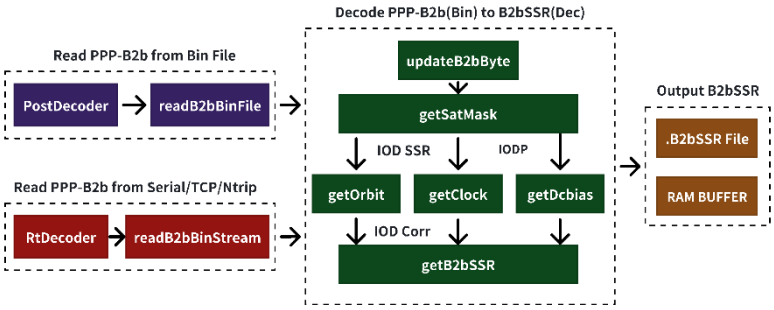


Fig 1. Schematic diagram of the RTKLIB-B2b decoder structure.

The positioning functionalities (both post-processing and real-time) represent a further application of the decoder module. They utilize the decoded PPP-B2b corrections and ephemeris data, along with code and phase observations, to compute the PPP solution.

The RTKLIB-B2b software supports two distinct decoding types: a post-processing decoder designed for handling PPP-B2b data files, and a real-time decoder tailored for processing actual data streams containing PPP-B2b corrections and broadcast ephemerides (including BDS CNAV1 and GPS LNAV).

In the post-processing mode, the operational logic of the decoder is driven by the timestamps of the observation data. A correction message is applied only if its timestamp is not later than (i.e., less than or equal to) the timestamp of the current observation data. If the correction timestamp is newer, the system must wait for subsequent observation data with a matching or later timestamp. This timestamp-based mechanism is also widely adopted in other libraries, such as RTKLIB for RTCM

processing (Takasu and Yasuda, n.d.), CLASLIB for CLAS processing (Motooka et al. 2019), and HASLIB for HAS processing (Zhang, Tu, and Lu 2024).

In the real-time mode, the decoder is running in an independent thread and driven by the data stream arrival events. Each time when new correction messages are received, relevant information are decoded and stored in temporary memories.

Fig 2 presents a block diagram of the RTKLIB-B2b positioning module, illustrating how it realizes the positioning function through three core components: data loading, mode-specific (post-processing/real-time) PPP-B2b decoding, and PPP computation based on RTKLIB.

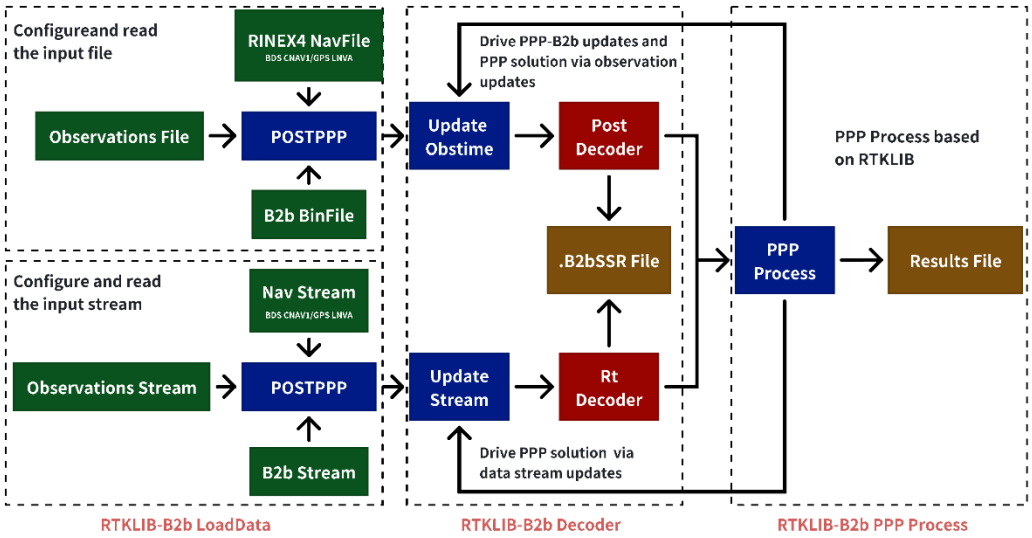


Fig 2. Block diagram of the RTKLIB-B2b positioning module structure.

(2) RTKLIB-B2b Software Compilation and Installation

Project Repository: https://github.com/UCAS-Liuchunbo/RTKLIB-B2b.git

Clone the repository using the address provided above. The structure of the cloned repository is as follows:

```
.
├── CMakeLists.txt
├── src/
├── include/
├── app/
├── build/
├── lib/
├── bin/
├── conf/
├── out/
├── manual/
└── example/
```

```
le/
```

- CMakeLists.txt: The script file used by CMake to build the project.
- src/: The directory contains the common source code files.
- include/: The directory contains the common header files.
- app/: The directory contains the functional-related source code files.
- build/: The directory intended for compilation.
- lib/: The directory intended for saving common static libraries.
- bin/: The directory intended for saving the executable programs.
- conf/: The directory contains configuration files.
- out/: The directory intended for saving result files.
- manual/: The directory contains this manual file.
- example/: The directory contains sample data and configuration files. Users will primarily work within this directory to run the examples.

The functional modules of this program are compiled together with the common source code library. For different operating systems, specific compilers are required for the compilation:

- Unix: The GCC compiler is necessary.
- Windows: Use GCC from the MinGW-w64 toolchain.

Executable files are obtained directly after compilation; no additional installation steps are required. For users interested in debugging or modifying the code, we recommend using the lightweight and free IDE, VS Code.

Note: The real-time modules utilize console functionality adapted from RTKLIB. However, due to limited support for the relevant APIs in Windows, the console feature cannot be enabled for real-time modules on Windows systems (though they can still run in non-console mode). The console mode allows convenient monitoring of real-time data stream transmission and decoding status, facilitating rapid troubleshooting. Therefore, we strongly recommend using a Unix-like system (e.g., Linux, macOS) to run this software for the optimal experience.

The demonstration environment is **Ubuntu 20.04 LTS**.

First, install the necessary tools (GCC compiler suite, CMake, and GDB) using your package manager.

```
sudo apt update
sudo apt install build-essential
sudo apt install cmake
sudo apt install gdb
```

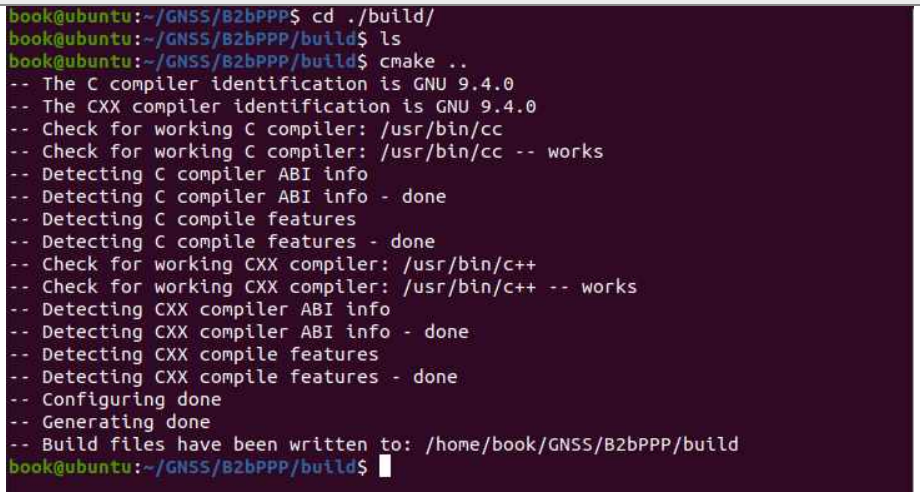
The versions used in the demonstration were:

```
book@ubuntu:/$ gcc --version
gcc (Ubuntu 9.4.0-1ubuntu1~20.04.2) 9.4.0
Copyright (C) 2019 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

book@ubuntu:/$ cmake --version
cmake version 3.16.3

CMake suite maintained and supported by Kitware (kitware.com/cmake).
book@ubuntu:/$ gdb --version
GNU gdb (Ubuntu 9.2-0ubuntu1~20.04.2) 9.2
Copyright (C) 2020 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <http://gnu.org/licenses/gpl.html>
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.
book@ubuntu:/$
```

Fig 3. Example figure (1)

Navigate into the build directory and generate the build system. (Note: Ensure the build directory is empty before running cmake. Clear all contents if necessary.)

```
cd ./build
cmake ..
```



```
book@ubuntu:~/GNSS/B2bPPP$ cd ./build/
book@ubuntu:~/GNSS/B2bPPP/build$ ls
book@ubuntu:~/GNSS/B2bPPP/build$ cmake ..
-- The C compiler identification is GNU 9.4.0
-- The CXX compiler identification is GNU 9.4.0
-- Check for working C compiler: /usr/bin/cc
-- Check for working C compiler: /usr/bin/cc -- works
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Detecting C compile features
-- Detecting C compile features - done
-- Check for working CXX compiler: /usr/bin/c++
-- Check for working CXX compiler: /usr/bin/c++ -- works
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done
-- Generating done
-- Build files have been written to: /home/book/GNSS/B2bPPP/build
book@ubuntu:~/GNSS/B2bPPP/build$
```

Fig 4. Example figure (2)

After cmake completes successfully, start the compilation:

```
make
```

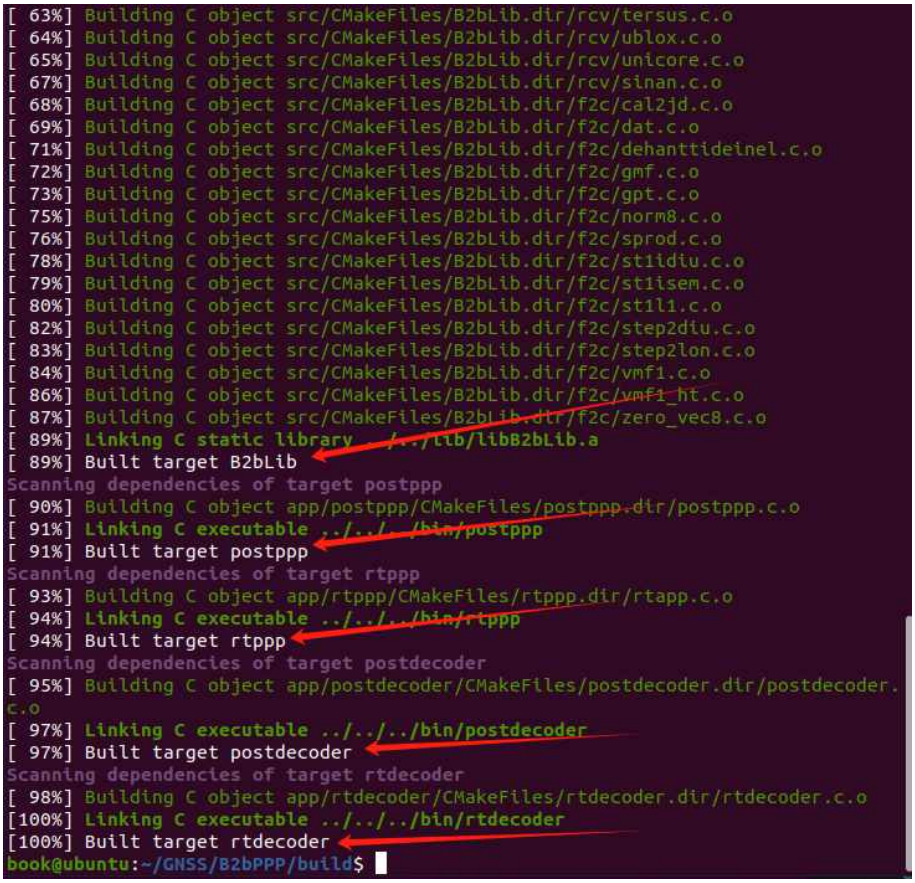


Fig 5. Example figure (3)

Upon successful compilation, navigate to the bin directory. You will find the compiled functional modules:

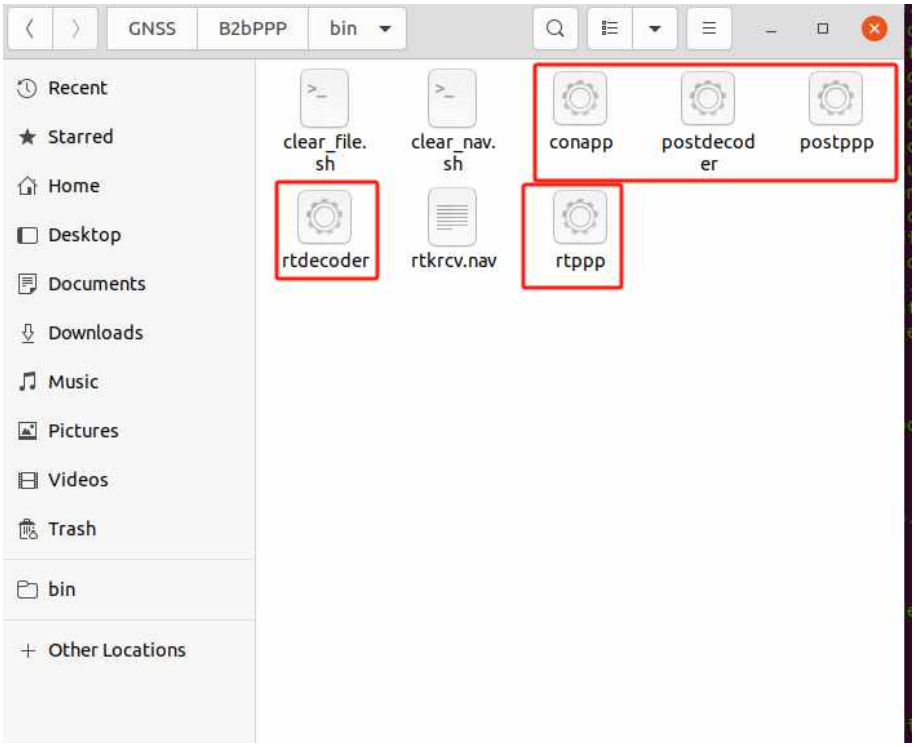


Fig 6. Example figure (4)

3. RTKLIB-B2b Decoder

Detailed specifications for the PPP-B2b formats from various manufacturers are provided in the appendix or attachments. In this section, we will introduce two different decoding methods and explain how to use the RTKLIB-B2b decoder.

(1) Post-processing Decoder

The RTKLIB-B2b post-processing decoder is designed for users who have already saved the binary data stream output from a receiver into a file, typically using serial communication software (such as RTKLIB's STRSVR). This decoder processes such files, decodes the information, and outputs it according to a specified format. Currently, this post-processing decoder supports the decoding of PPP-B2b information in both SINO and UNICORE formats.

a. Usage

The example for this module is located in the RTKLIB-B2b/example/postdecoder directory. After compilation is complete, users can replace the postdecoder program in this directory with their newly compiled version for testing. The post-processing decoder, postdecoder, is launched via command-line arguments.

```
postdecoder [-U] [-S] [-in input_B2bfile] [-out out_Resultfile]
```

b. Options

-U : Specifies decoding in UNICORE format.
-S : Specifies decoding in SINO format.
-in : Specifies the path to the input B2b Binfile.
-out : Specifies the path where the decoded B2b result file will be stored.

c. Example

```
./postdecoder -U -in ./Unicore_2024360.B2bBin -out ./UNICORE_2024360.B2bSSR
```

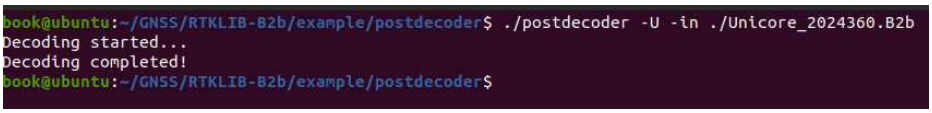
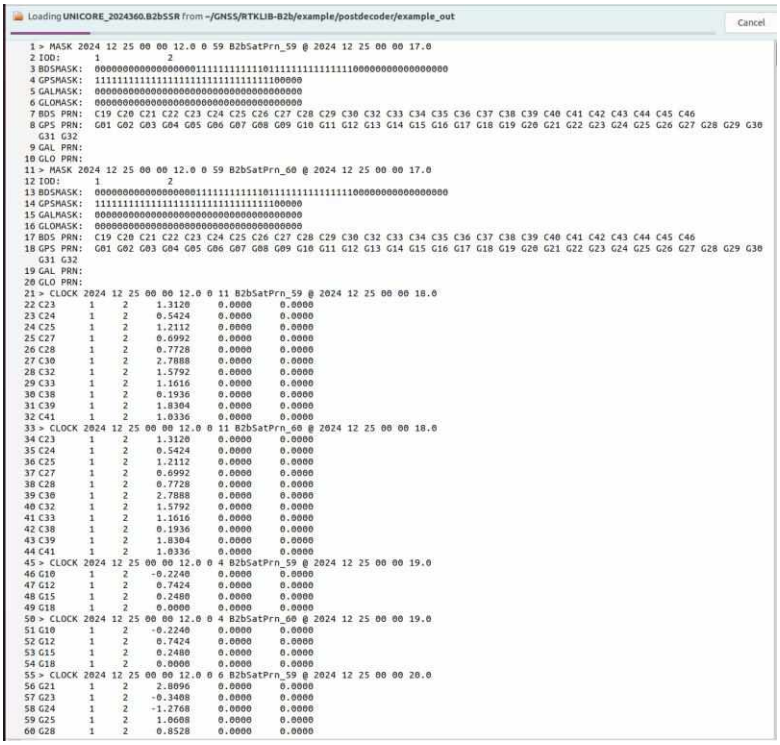


Fig 7. Example of running RTKLIB-B2b-PostDecoder in the command line

After decoding is complete, a text result file will be generated in the specified output directory. The format description for this file will be provided in the final part of this section.



```
Loading UNICORE_2024360.B2bSSR from ~/GNSS/RTKLIB-B2b/example/postdecoder/example_out                    Cancel

 1 > MASK 2024 12 25 00 00 12.0 0 59 B2bSatPrn_59 @ 2024 12 25 00 00 17.0
 2 IOD:      1            2
 3 BDSMASK:  000000000000000000111111111111011111111111111100000000000000000
 4 GPSMASK:  111111111111111111111111111111100000
 5 GALMASK:  00000000000000000000000000000000000
 6 GLOMASK:  00000000000000000000000000000000000
 7 BDS PRN:  C19 C20 C21 C22 C23 C24 C25 C26 C27 C28 C29 C30 C32 C33 C34 C35 C36 C37 C38 C39 C40 C41 C42 C43 C44 C45 C46
 8 GPS PRN:  G01 G02 G03 G04 G05 G06 G07 G08 G09 G10 G11 G12 G13 G14 G15 G16 G17 G18 G19 G20 G21 G22 G23 G24 G25 G26 G27 G28 G29 G30
   G31 G32
 9 GAL PRN:
10 GLO PRN:
11 > MASK 2024 12 25 00 00 12.0 0 59 B2bSatPrn_60 @ 2024 12 25 00 00 17.0
12 IOD:      1            2
13 BDSMASK:  000000000000000000111111111111011111111111111100000000000000000
14 GPSMASK:  111111111111111111111111111111100000
15 GALMASK:  00000000000000000000000000000000000
16 GLOMASK:  00000000000000000000000000000000000
17 BDS PRN:  C19 C20 C21 C22 C23 C24 C25 C26 C27 C28 C29 C30 C32 C33 C34 C35 C36 C37 C38 C39 C40 C41 C42 C43 C44 C45 C46
18 GPS PRN:  G01 G02 G03 G04 G05 G06 G07 G08 G09 G10 G11 G12 G13 G14 G15 G16 G17 G18 G19 G20 G21 G22 G23 G24 G25 G26 G27 G28 G29 G30
   G31 G32
19 GAL PRN:
20 GLO PRN:
21 > CLOCK 2024 12 25 00 00 12.0 0 11 B2bSatPrn_59 @ 2024 12 25 00 00 18.0
22 C23      1     2     1.3120     0.0000     0.0000
23 C24      1     2     0.5424     0.0000     0.0000
24 C25      1     2     1.2112     0.0000     0.0000
25 C27      1     2     0.6992     0.0000     0.0000
26 C28      1     2     0.7728     0.0000     0.0000
27 C30      1     2     2.7888     0.0000     0.0000
28 C32      1     2     1.5792     0.0000     0.0000
29 C33      1     2     1.1616     0.0000     0.0000
30 C38      1     2     0.1936     0.0000     0.0000
31 C39      1     2     1.8304     0.0000     0.0000
32 C41      1     2     1.0336     0.0000     0.0000
33 > CLOCK 2024 12 25 00 00 12.0 0 11 B2bSatPrn_60 @ 2024 12 25 00 00 18.0
34 C23      1     2     1.3120     0.0000     0.0000
35 C24      1     2     0.5424     0.0000     0.0000
36 C25      1     2     1.2112     0.0000     0.0000
37 C27      1     2     0.6992     0.0000     0.0000
38 C28      1     2     0.7728     0.0000     0.0000
39 C30      1     2     2.7888     0.0000     0.0000
40 C32      1     2     1.5792     0.0000     0.0000
41 C33      1     2     1.1616     0.0000     0.0000
42 C38      1     2     0.1936     0.0000     0.0000
43 C39      1     2     1.8304     0.0000     0.0000
44 C41      1     2     1.0336     0.0000     0.0000
45 > CLOCK 2024 12 25 00 00 12.0 0 4 B2bSatPrn_59 @ 2024 12 25 00 00 19.0
46 G10      1     2    -0.2240     0.0000     0.0000
47 G12      1     2     0.7424     0.0000     0.0000
48 G15      1     2     0.2480     0.0000     0.0000
49 G18      1     2     0.0000     0.0000     0.0000
50 > CLOCK 2024 12 25 00 00 12.0 0 4 B2bSatPrn_60 @ 2024 12 25 00 00 19.0
51 G10      1     2    -0.2240     0.0000     0.0000
52 G12      1     2     0.7424     0.0000     0.0000
53 G15      1     2     0.2480     0.0000     0.0000
54 G18      1     2     0.0000     0.0000     0.0000
55 > CLOCK 2024 12 25 00 00 12.0 0 6 B2bSatPrn_59 @ 2024 12 25 00 00 20.0
56 G21      1     2     2.8096     0.0000     0.0000
57 G23      1     2    -0.3408     0.0000     0.0000
58 G24      1     2    -1.2768     0.0000     0.0000
59 G25      1     2     1.0608     0.0000     0.0000
60 G28      1     2     0.8528     0.0000     0.0000
```

Fig 8. Example of .B2bSSR file obtained from decoding.

(2) Real-time Decoder

The RTKLIB-B2b real-time decoder allows users to decode the binary data stream output from a receiver in real-time via methods such as Serial, TCP/IP, Ntrip, etc., and output the results in a specified format. This real-time decoder currently supports the decoding of GPS LNAV, BDS CNAV1, and PPP-B2b information in both SINO and UNICORE formats. It also supports a playback mode (following RTKLIB playback rules).

a. Usage

The example for this module is located in the RTKLIB-B2b/test/rtdecoder directory. After compilation is complete, users can replace the rtdecoder program in this directory with their newly compiled version for testing. The real-time decoder, rtdecoder, is launched using a configuration file passed via command-line arguments. (When using a serial port for data reception, ensure the user has the necessary serial port access permissions.)

```
rtdecoder [-o config_file] [-s] [-nc]
```

b. Options

-o: Specifies the path to the configuration file.

-s: Specifies launching in console mode (this option is unavailable on Windows

systems).

-nc: Specifies launching in non-console mode.

c. Console Commands

Console commands are instructions used to interact with the program via the command line after launching in console mode (-s) to view the current real-time decoding status. The format remains consistent with RTKLIB (demo5), with modifications specific to B2b. Currently available commands are:

- sta: Prints the current status information of the program decoding, including basic software info, and counts of ephemeris and PPP-B2b messages.
- nav: Prints the currently decoded ephemeris information.
- ssr: Prints the currently decoded PPP-B2b information.
- stream: Views the current status information of the data streams.
- shutdown:Terminates the current process (when receiving data via Serial or TCP/IP, please ensure you use this command to properly terminate the process).

d. Example

Note that in this example, the configuration file is pre-filled with a test account for receiving data via Ntrip. Test accounts may experience instability. Therefore, if you have your own PPP-B2b receiver, we recommend replacing the configuration details with your own.

Enter the example directory and run the following command to start.

```
./rtdecoder -o ./rtdecoder_mycaster_sino.conf -s
```

Fig 9. Example of the printout after running the 'sta' command in the RTKLIB-B2b real-time decoder (console mode).

```
** B2bPPP ver.v1.0 20250305 console (h:help) **
rtk server start
B2bPPP> stream

Stream       Type     Fmt   S    In-byte  In-bps   Out-byte Out-bps Path                    Message
input stream1 ntripc   sino  C       1001    3968          0       0 [redacted]@1.94.195.74: 1.94.195.74/SINO_Nav
input stream2 ntripc   sino  C       1178    7142          0       0 [redacted]@1.94.195.74: 1.94.195.74/SINO_B2b
log stream1  file     -     C          0       0       1001    3970 ./Bas_%Y%m%d%h%M.log::T:
log stream2  file     -     C          0       0       1178    2517 ./Cor_%Y%m%d%h%M.log::T:
B2bPPP>
```

Fig 10. Example of the printout after running the 'stream' command in the RTKLIB-B2b real-time decoder (console mode).

```
B2bPPP server state          : run
processing cycle (ms)        : 10
# of input data stream_1     : nav(63),B2bssr(0),err(0)
# of input data stream_2     : nav(0),B2bssr(3518),err(0)
# of nav messages (sino)     : gps_lnav(1158), bds_cnav1(1100)
# of B2bssr messages (sino) : mask(48), eph_ura(191), diff_cbia(192), clk(1157), others(1885)
B2bPPP> ssr

SAT   S UDI IODSSR URA                  T0   D0-A   D0-C   D0-R       C0     C1     C2
G05  OK  48   1     39 2025/03/07 17:05:41  0.277 -0.640 -0.742    2.360  0.000 0.0000
G10  OK  48   1     39 2025/03/07 17:05:41  0.198  1.242 -0.173   -0.542  0.000 0.0000
G13  OK  48   1     39 2025/03/07 17:05:41 -0.061  0.858  0.307   -0.576  0.000 0.0000
G15  OK  48   1     39 2025/03/07 17:05:41  0.243  0.160 -0.691    0.376  0.000 0.0000
G18  OK  48   1     39 2025/03/07 17:05:41  0.181 -0.947 -0.186    0.000  0.000 0.0000
G23  OK  48   1     39 2025/03/07 17:05:41  0.016  0.115 -0.064    0.392  0.000 0.0000
G24  OK  48   1     39 2025/03/07 17:05:41  0.088 -1.478  0.550   -0.742  0.000 0.0000
G32  OK  48   1     39 2025/03/07 17:05:41  0.107  0.858 -0.454   -1.226  0.000 0.0000
C19  OK  48   1     31 2025/03/07 17:05:41 -0.045  0.045 -0.038   -0.029  0.000 0.0000
C20  OK  48   1     37 2025/03/07 17:01:41  0.011  0.000 -0.090    0.262  0.000 0.0000
C21  OK  48   1     31 2025/03/07 17:05:41  0.005 -0.026  0.026    0.400  0.000 0.0000
C22  OK  48   1     36 2025/03/07 17:05:41  0.021 -0.058  0.000   -0.544  0.000 0.0000
C29  OK  48   1     39 2025/03/07 16:47:17  0.027 -0.064 -0.013   -0.486  0.000 0.0000
C35  OK  48   1     31 2025/03/07 17:05:41  0.018 -0.038 -0.109    0.256  0.000 0.0000
C36  OK  48   1     31 2025/03/07 17:05:41  0.022  0.000  0.115    0.278  0.000 0.0000
C38  OK  48   1     31 2025/03/07 17:05:41 -0.027  0.083 -0.122   -0.003  0.000 0.0000
C39  OK  48   1     31 2025/03/07 17:05:41  0.018  0.032  0.531   -0.357  0.000 0.0000
C44  OK  48   1     31 2025/03/07 17:05:41  0.038 -0.038 -0.115   -0.190  0.000 0.0000
C45  OK  48   1     31 2025/03/07 17:05:41  0.005  0.013  0.077    0.840  0.000 0.0000
B2bPPP>
```

Fig 11. Example of the printout after running the 'ssr' command in the RTKLIB-B2b real-time decoder (console mode).

```
C44  OK  48   1     31 2025/03/07 17:05:41  0.038 -0.038 -0.115   -0.190  0.000 0.0000
C45  OK  48   1     31 2025/03/07 17:05:41  0.005  0.013  0.077    0.840  0.000 0.0000
B2bPPP> nav

SAT   S IOD IOC FRQ A/A SVH                 Toe                 Toc             Ttr/Tof L2C L2P
G01   - 189 957   0   0 000 2025/03/07 10:00:00 2025/03/07 10:00:00 2025/03/07 16:46:43   0   0
G02   -  40  40   0   0 000 2025/03/07 08:00:00 2025/03/07 08:00:00 2025/03/07 16:46:44   0   0
G03   -  32  32   0   0 000 2025/03/07 10:00:00 2025/03/07 10:00:00 2025/03/07 16:46:45   0   0
G04   - 133 645   0   0 000 2025/03/07 12:00:00 2025/03/07 12:00:00 2025/03/07 16:46:46   0   0
G05  OK  96  96   0   0 000 2025/03/07 18:00:00 2025/03/07 18:00:00 2025/03/07 16:46:47   0   0
G06   -  22  22   0   0 000 2025/03/07 14:00:00 2025/03/07 14:00:00 2025/03/07 16:46:48   0   0
G07   - 109 109   0   0 000 2025/03/07 14:00:00 2025/03/07 14:00:00 2025/03/07 16:46:49   0   0
G08   -  53  53   0   0 000 2025/03/07 08:00:00 2025/03/07 08:00:00 2025/03/07 16:46:50   0   0
G09   -  12  12   0   0 000 2025/03/07 14:00:00 2025/03/07 14:00:00 2025/03/07 16:46:51   0   0
G10  OK  90  90   0   0 000 2025/03/07 18:00:00 2025/03/07 18:00:00 2025/03/07 16:46:52   0   0
G11  OK  67  67   0   0 000 2025/03/07 16:00:00 2025/03/07 16:00:00 2025/03/07 16:46:53   0   0
G12  OK  21  21   0   0 000 2025/03/07 17:59:44 2025/03/07 17:59:44 2025/03/07 16:59:50   0   0
G13  OK  53  53   0   0 000 2025/03/07 18:00:00 2025/03/07 18:00:00 2025/03/07 16:46:19   0   0
G14   - 206 206   0   0 000 2025/03/07 10:00:00 2025/03/07 10:00:00 2025/03/07 16:46:20   0   0
G15  OK  12  12   0   0 000 2025/03/07 17:59:44 2025/03/07 17:59:44 2025/03/07 16:46:21   0   0
G16   -  45  45   0   0 000 2025/03/07 05:59:44 2025/03/07 05:59:44 2025/03/07 16:46:22   0   0
G17   -  37  37   0   0 000 2025/03/07 12:00:00 2025/03/07 12:00:00 2025/03/07 16:46:23   0   0
G18  OK  32 544   0   0 000 2025/03/07 18:00:00 2025/03/07 18:00:00 2025/03/07 16:46:24   0   0
G19   -  18  18   0   0 000 2025/03/07 14:00:00 2025/03/07 14:00:00 2025/03/07 16:46:25   0   0
G20  OK  33  33   0   0 000 2025/03/07 16:00:00 2025/03/07 16:00:00 2025/03/07 16:46:26   0   0
G21   - 130 130   0   0 000 2025/03/07 08:00:00 2025/03/07 08:00:00 2025/03/07 16:46:27   0   0
G22   -  71  71   0   0 000 2025/03/07 12:00:00 2025/03/07 12:00:00 2025/03/07 16:46:28   0   0
G23  OK  36 804   0   0 000 2025/03/07 18:00:00 2025/03/07 18:00:00 2025/03/07 16:46:29   0   0
G24  OK  49  49   0   0 000 2025/03/07 18:00:00 2025/03/07 18:00:00 2025/03/07 16:46:30   0   0
G25   -  73  73   0   0 000 2025/03/07 14:00:00 2025/03/07 14:00:00 2025/03/07 16:46:31   0   0
G26   -  18  18   0   0 000 2025/03/07 03:59:44 2025/03/07 03:59:44 2025/03/07 16:46:32   0   0
G27   -  20  20   0   0 000 2025/03/07 06:00:00 2025/03/07 06:00:00 2025/03/07 16:46:33   0   0
G28   -  70 582   0   0 000 2025/03/07 02:00:00 2025/03/07 02:00:00 2025/03/07 16:46:34   0   0
G29  OK  44  44   0   0 000 2025/03/07 18:00:00 2025/03/07 18:00:00 2025/03/07 16:46:35   0   0
G30  OK  81  81   0   0 000 2025/03/07 16:00:00 2025/03/07 16:00:00 2025/03/07 16:46:36   0   0
G31   -  22  22   0   0 000 2025/03/07 04:00:00 2025/03/07 04:00:00 2025/03/07 16:46:37   0   0
G32  OK  52  52   0   0 000 2025/03/07 18:00:00 2025/03/07 18:00:00 2025/03/07 16:47:50   0   0
C19  OK  25  25   0   0 000 2025/03/07 17:00:14 2025/03/07 17:00:14 2025/03/07 17:00:33   0   0
C20  OK  24  24   0   0 000 2025/03/07 16:00:14 2025/03/07 16:00:14 2025/03/07 16:46:27   0   0
C21  OK  25  25   0   0 000 2025/03/07 17:00:14 2025/03/07 17:00:14 2025/03/07 17:00:57   0   0
C22  OK  25  25   0   0 000 2025/03/07 17:00:14 2025/03/07 17:00:14 2025/03/07 17:00:36   0   0
C27   -  20  20   0   0 000 2025/03/07 12:00:14 2025/03/07 12:00:14 2025/03/07 16:46:30   0   0
C29  OK  24  24   0   0 000 2025/03/07 16:00:14 2025/03/07 16:00:14 2025/03/07 16:46:31   0   0
C30   -  22  22   0   0 000 2025/03/07 14:00:14 2025/03/07 14:00:14 2025/03/07 16:46:32   0   0
C32   -  21  21   0   0 000 2025/03/07 13:00:14 2025/03/07 13:00:14 2025/03/07 16:46:33   0   0
```

Fig 12. Example of the printout after running the 'nav' command in the RTKLIB-B2b real-time decoder (console mode).

```
> MASK 2025 03 07 16 46 36.0 0 59 B2bSatPrn_59 @ 2025 03 07 16 46 41.1
IOD:      1            2
BDSMASK:  000000000000000000111111111111011111111111111000000000000000000
GPSMASK:  111111111111111111111111111111100000
GALMASK:  000000000000000000000000000000000000
GLOMASK:  000000000000000000000000000000000000
BDS PRN:  C19 C20 C21 C22 C23 C24 C25 C26 C27 C28 C29 C30 C32 C33 C34 C35 C36 C37 C38 C39 C40 C41 C42 C43 C44 C45 C46
GPS PRN:  G01 G02 G03 G04 G05 G06 G07 G08 G09 G10 G11 G12 G13 G14 G15 G16 G17 G18 G19 G20 G21 G22 G23 G24 G25 G26 G27 G28 G29 G30 G31 G32
GAL PRN:
GLO PRN:
> MASK 2025 03 07 16 46 36.0 0 59 B2bSatPrn_60 @ 2025 03 07 16 46 41.1
IOD:      1            2
BDSMASK:  000000000000000000111111111111011111111111111000000000000000000
GPSMASK:  111111111111111111111111111111100000
GALMASK:  000000000000000000000000000000000000
GLOMASK:  000000000000000000000000000000000000
BDS PRN:  C19 C20 C21 C22 C23 C24 C25 C26 C27 C28 C29 C30 C32 C33 C34 C35 C36 C37 C38 C39 C40 C41 C42 C43 C44 C45 C46
GPS PRN:  G01 G02 G03 G04 G05 G06 G07 G08 G09 G10 G11 G12 G13 G14 G15 G16 G17 G18 G19 G20 G21 G22 G23 G24 G25 G26 G27 G28 G29 G30 G31 G32
GAL PRN:
GLO PRN:
> CLOCK 2025 03 07 16 46 36.0 0 11 B2bSatPrn_59 @ 2025 03 07 16 46 42.1
C19      1     2    -0.1200     0.0000     0.0000
C20      1     2     0.2272     0.0000     0.0000
C21      1     2     0.3184     0.0000     0.0000
C22      1     2    -0.1232     0.0000     0.0000
C29      1     2    -0.4720     0.0000     0.0000
C34      1     2     0.1392     0.0000     0.0000
C35      1     2    -0.1200     0.0000     0.0000
C36      1     2     0.0656     0.0000     0.0000
C37      1     2    -0.0720     0.0000     0.0000
C38      1     2     0.0672     0.0000     0.0000
C39      1     2     0.1984     0.0000     0.0000
> CLOCK 2025 03 07 16 46 36.0 0 11 B2bSatPrn_60 @ 2025 03 07 16 46 42.1
C19      1     2    -0.1200     0.0000     0.0000
C20      1     2     0.2272     0.0000     0.0000
C21      1     2     0.3184     0.0000     0.0000
C22      1     2    -0.1232     0.0000     0.0000
C29      1     2    -0.4720     0.0000     0.0000
C34      1     2     0.1392     0.0000     0.0000
C35      1     2    -0.1200     0.0000     0.0000
C36      1     2     0.0656     0.0000     0.0000
C37      1     2    -0.0720     0.0000     0.0000
```

Fig 13. Example of the resulting .B2bSSR file obtained from the RTKLIB-B2b real-time decoder (console mode).

(3) Decoding result format description

To balance user convenience with software compatibility, the RTKLIB-B2b decoder's output format was developed based on the IGS SSR format used by the BNC software(“BKG Ntrip Client - BNC,” n.d.). Due to numerous differences between PPP-B2b and IGS SSR in aspects like message types and parameter types, we have extended

the original format to create the specific PPP-B2b output format for the RTKLIB-B2b decoder. The detailed format description follows.

a. Epoch Record

- The special character > marks the beginning of an "Epoch Record".
- B2bSSR Message/Topic Descriptor, Valid descriptors currently include

MASK, ORBIT_URAI, DIFF_CODE_BIAS, CLOCK

These correspond respectively to PPP-B2b Information Types 1~4:

Satellite mask、Orbit corrections and User Range Accuracy (URA) index、Differential Code Bias (DCB) corrections、Clock corrections

- Message Epoch Reference Time - Year (GPS Time)
- Message Epoch Reference Time - Month (GPS Time)
- Message Epoch Reference Time - Day (GPS Time)
- Message Epoch Reference Time - Hour (GPS Time)
- Message Epoch Reference Time - Minute (GPS Time)
- Message Epoch Reference Time - Second (GPS Time)
- B2bSSR Message Update Interval (seconds):

This value might be inaccurate during initial reception and decoding. This is because, firstly, calculating this interval requires at least two messages of the same type, and secondly, it can be affected by the receiver's own startup process and the completeness of the broadcast B2b messages. In such cases, this parameter will be set to 0.

For MASK messages, if messages from multiple B2b satellites are received concurrently (e.g., PRN60 received after PRN59), the update interval field in the MASK message for PRN60 will be set to 0.

If accurate usage of this parameter is essential, it is recommended to re-iterate through the entire output file and check this value after the initial processing is complete.

- Number of Subsequent Records in this Block:

For MASK messages, this indicates the number of satellites with a mask value of 1. For other message types, this indicates the count of valid decoded information items. Note that this count of valid items is not necessarily the same as the "number of satellites" described in the interface documents, because the "number of satellites"

mentioned in interface documents might still include satellites that have been excluded due to poor accuracy. Therefore, the count here refers to the number of usable satellites whose decoded information falls within the specified accuracy threshold.

- Identifier of the Satellite Broadcasting PPP-B2b: Currently includes C59, C60, C61(62).
- Receiver Reception Time: Indicated by an @ symbol following the broadcasting satellite identifier, followed by:
  - Receiver Reception Time - Year (GPS Time)
  - Receiver Reception Time - Month (GPS Time)
  - Receiver Reception Time - Day (GPS Time)
  - Receiver Reception Time - Hour (GPS Time)
  - Receiver Reception Time - Minute (GPS Time)
  - Receiver Reception Time - Second (GPS Time)

b. Satellite Mask 'MASK' Example

```
> MASK 2025 03 07 16 59 24.0 48 59 B2bSatPrn_59 @ 2025 03 07 16 59 29.2
IOD:        1                2
BDSMASK:   0000000000000000001111111111110111111111111110000000000000000
GPSMASK:   11111111111111111111111111111111100000
GALMASK:   00000000000000000000000000000000000
GLOMASK:   00000000000000000000000000000000000
BDS PRN:   C19 C20 C21 C22 C23 C24 C25 C26 C27 C28 C29 C30 C32 C33 C34 C35 C36 C37 C38 C39 C40 C41
C42 C43 C44 C45 C46
GPS PRN:   G01 G02 G03 G04 G05 G06 G07 G08 G09 G10 G11 G12 G13 G14 G15 G16 G17 G18 G19 G20 G21 G22
G23 G24 G25 G26 G27 G28 G29 G30 G31 G32
GAL PRN:
GLO PRN:
```

Records in this block provide the following satellite-specific information (by row):

- IOD: Issue of Data SSR (IODSSR) and Issue of Data PPP (IODP) respectively. IODSSR corresponds to the IODSSR of each message, while IODP corresponds to the IODP in Message Type 4 (Clock).
- BDS MASK: BDS mask represented in ASCII format.
- GPS MASK: GPS mask represented in ASCII format.
- GAL MASK: Galileo mask represented in ASCII format.
- GLO MASK: GLONASS mask represented in ASCII format.

- BDS PRN: List of all BDS satellites with MASK=1 (SLOT number converted to PRN).
- GPS PRN: List of all GPS satellites with MASK=1 (SLOT number converted to PRN).
- GAL PRN: List of all Galileo satellites with MASK=1 (SLOT number converted to PRN).
- GLO PRN: List of all GLONASS satellites with MASK=1 (SLOT number converted to PRN).

c. Orbit Corrections and URAI 'ORBIT_URAI' Example

```
> ORBIT_URAI 2025 03 07 16 59 17.0 0 6 B2bSatPrn_59 @ 2025 03 07 16 59 41.2
C36    1    24    3    31    0.0224    0.0384    0.0768    0.0000    0.0000    0.0000
C37    1    24    5    31   -0.0528   -0.0896    0.0000    0.0000    0.0000    0.0000
C38    1    24    1    31    0.0064    0.0640   -0.0768    0.0000    0.0000    0.0000
C39    1    24    1    31    0.0240   -0.0832    0.7872    0.0000    0.0000    0.0000
C44    1    24    7    31    0.0032   -0.0192   -0.1088    0.0000    0.0000    0.0000
C45    1    24    1    31    0.0096    0.0512    0.0448    0.0000    0.0000    0.0000
```

Records in this block provide the following satellite-specific information (by column):

- Satellite PRN number (G=GPS, R=GLONASS, E=Galileo, C=BDS).
- IODSSR number for this message, corresponding to IODSSR in the MASK message.
- IODN number for this message, corresponding to IODC in the broadcast ephemeris.
- IOD Corr for this message, corresponding to IOD Corr in Message Type 4 (Clock).
- User Range Accuracy Index (URAI) for this message.
- Broadcast ephemeris orbit correction - Radial component [meters].
- Broadcast ephemeris orbit correction - Along component [meters].
- Broadcast ephemeris orbit correction - Cross component [meters].
- Broadcast ephemeris orbit correction velocity - Radial component [mm/s].
- Broadcast ephemeris orbit correction velocity - Along component [mm/s].
- Broadcast ephemeris orbit correction velocity - Along component [mm/s].

d. Differential Code Bias Corrections 'DIFF_CODE_BIAS' Example

```
> DIFF_CODE_BIAS 2025 03 07 16 58 38.0 0 3 B2bSatPrn_60 @ 2025 03 07 16 59 39.1
C39    1    7    1P   2.8900    7I -1.5810    7Q -1.0880    2I   1.9720    1D   2.6860    5D -2.7880    5P -1.9890
C44    1    7    1P -0.6460    7I -1.4450    7Q -0.9180    2I -0.3570    1D -0.8500    5D -2.0910    5P -1.3260
C45    1    7    1P   7.8200    7I -2.3120    7Q -1.8190    2I   5.7120    1D   7.5650    5D -3.2640    5P -2.4990
```

Records in this block provide the following satellite-specific information (by column):

- Satellite PRN number (G=GPS, R=GLONASS, E=Galileo, C=BDS).
- IODSSR number for this message, corresponding to IODSSR in the MASK message.
- Signal and tracking mode indicators (e.g., 1C, 5Q, indicating frequency band and modulation).
- Differential Code Bias (DCB) value for the corresponding signal [meters], arranged in "Indicator-BiasValue" pairs.

e. Clock Corrections 'CLOCK' Example

```
> CLOCK 2025 03 07 17 09 24.0 6 9 B2bSatPrn_60 @ 2025 03 07 17 09 30.1
C19    1    2    -0.0352    0.0000    0.0000
C21    1    2     0.4064    0.0000    0.0000
C22    1    2    -0.5616    0.0000    0.0000
C34    1    2     0.2752    0.0000    0.0000
C35    1    2     0.2592    0.0000    0.0000
C36    1    2     0.2752    0.0000    0.0000
C37    1    2    -0.1328    0.0000    0.0000
C38    1    2    -0.0032    0.0000    0.0000
C39    1    2    -0.3728    0.0000    0.0000
```

Records in this block provide the following satellite-specific information (by column):

- Satellite PRN number (G=GPS, R=GLONASS, E=Galileo, C=BDS).
- IOD SSR
- IODP
- Clock correction polynomial coefficient C0 [meters].
- Clock correction polynomial coefficient C1 [mm/s].
- Clock correction polynomial coefficient C2 [mm/s²].

4. RTKLIB-B2b Positioning

(1) RTKLIB-B2b Post-processing PPP Positioning

The RTKLIB-B2b post-processing positioning functional module (Abbreviated as

RTKLIB-B2b PostPPP) is an extension built upon the RTKLIB-B2b post-processing decoder and RTKLIB's rnx2rtkp module. By using post-saved binary PPP-B2b files combined with broadcast ephemeris in RINEX 4 format (Montenbruck & Steigenberger, 2022), users can perform PPP positioning for observation stations located within the PPP-B2b coverage area.

a. Usage

The example for this module is located in the RTKLIB-B2b/example/postppp directory. After compilation is complete, users can replace the postppp program in this directory with their newly compiled version for testing. The post-processing positioning module, postppp, is launched using a configuration file passed via command-line arguments. To facilitate batch processing, it also allows users to override the input file, output file, satellite systems for processing, and processing mode via command-line parameters.

```
postppp [-o config_file] [-in input_file] [-out output_file] [-s system] [-m mode]
```

b. Options

-o: Specifies the path to the configuration file.

-in: Passes the input file path, overriding the input file specified in the configuration file (optional).

-out: Specifies the output file path, overriding the output file specified in the configuration file (optional).

-s: Specifies the satellite system(s) to be included in the solution, overriding the system(s) specified in the configuration file (optional).

-m: Specifies the processing mode, overriding the mode specified in the configuration file (optional).

c. Example

The data required for this post-processing example is publicly available from the Zenodo archive at: https://doi.org/10.5281/zenodo.15110885

This example demonstrates performing RTKLIB-B2b post-processing positioning for stations WUH2, GAMG, and ULAB using data from DOY 065,066 2025. Navigate to the example directory (typically RTKLIB-B2b/example/postppp) in your terminal and execute the following commands:

```
./postppp -o ./conf/postppp_25065_wuh2.conf
```

```
./postppp -o ./conf/postppp_25065_gamg.conf
./postppp -o ./conf/postppp_25065_ulab.conf
./postppp -o ./conf/postppp_25066_wuh2.conf
./postppp -o ./conf/postppp_25066_gamg.conf
./postppp -o ./conf/postppp_25066_ulab.conf
```

During this process, the command-line terminal should display warning messages related to PPP-B2b matching and information about the processing progress.

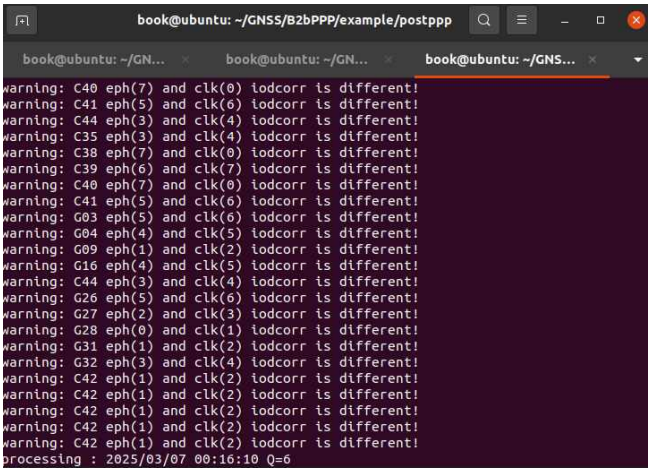


Fig 14. RTKLIB-B2b PostPPP Execution Example.

After processing is complete, users should obtain three files for each station: a file ending in .pos containing the positioning results, a file ending in .stat containing the processing status information, and a .pos file containing _events in its name, which stores results from epochs judged as abnormal during processing (under normal circumstances, this file should not contain any positioning results).

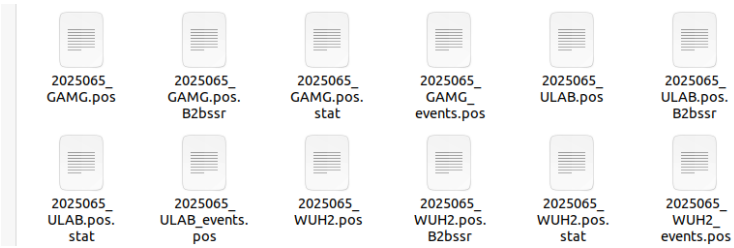


Fig 15. Generated Output Files (DOY 065)

Users can directly plot the positioning results using RTKLIB's rtkplot (The true value coordinates are derived from the IGS weekly SNX file of the corresponding week):

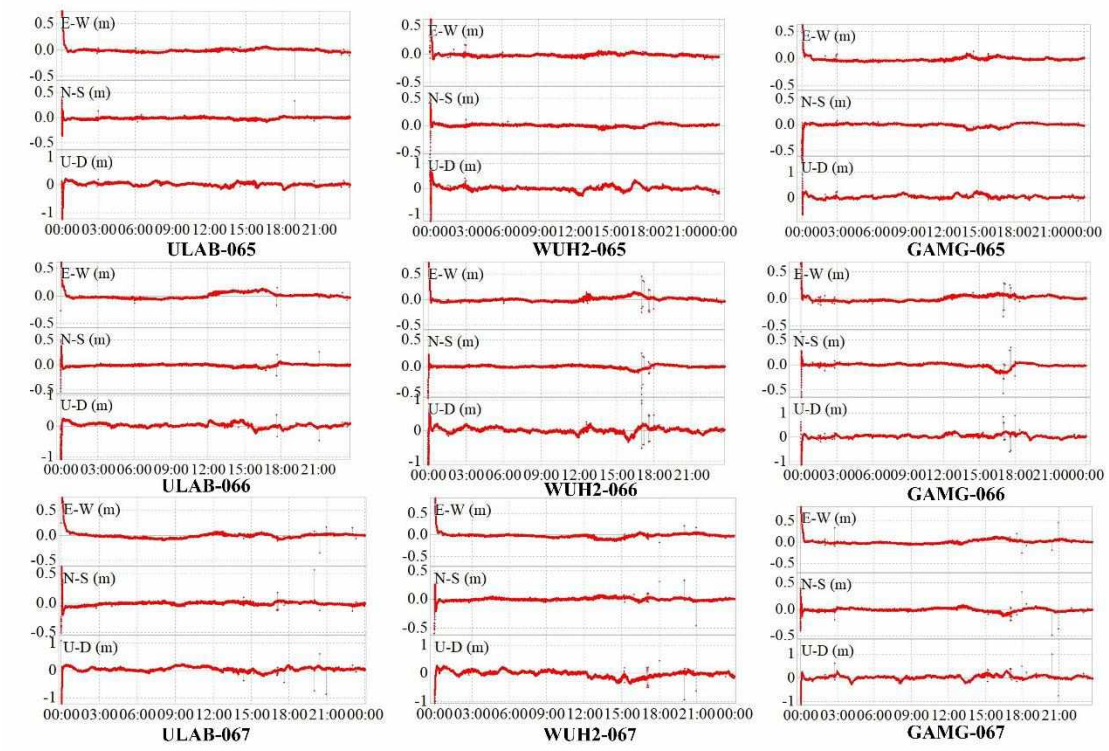


Fig 16. Example results Plotted using RTKLIB's rtkplot.

(2) RTKLIB-B2b Real-time PPP Positioning

The RTKLIB-B2b real-time positioning functional module represents an extension built upon the RTKLIB-B2b real-time decoder and incorporates functionalities adapted from RTKLIB's rtkrcv module. Utilizing real-time data streams containing PPP-B2b corrections and broadcast ephemerides, users can perform real-time Precise Point Positioning (PPP) for observation stations situated within the PPP-B2b service coverage area.

a. Usage

Example usage scenarios and files for this module are located in the RTKLIB-B2b/ example/rtppp directory. After successful compilation of the software suite, users should ensure the compiled rtppp executable is present in this directory (potentially replacing any placeholder or older version) before conducting tests.

```
rtdecoder [-o config_file] [-s] [-nc]
```

b. Options

-o: Specifies the full path to the configuration file.

-s: Launches the program in console mode, allowing interactive commands. (This option is unavailable on Windows systems).

-nc: Launches the program in non-console (background) mode.

-t: Specifies the debug trace level (typically written to the .trace file). If debugging is not required, a setting of 0 (off) is recommended, as trace output files can often become very large.

-r: Specifies the level of detail for the solution status output (typically written to the .stat file). A setting of 2 is recommended, as analyzing the information output at this level often helps diagnose most processing problems or issues.

c. Console Commands

When operating in console mode (-s), users can interact with the running rtppp process using specific commands entered at the prompt. These commands allow monitoring of the real-time processing status. The command structure maintains consistency with RTKLIB (demo5), incorporating further modifications tailored for PPP-B2b functionality.

The currently available console commands are:

sta : Prints current status information, including basic software version details, the number of valid ephemeris entries, and the count of received/processed PPP-B2b messages.

obs : Prints information about the currently processed observations.

nav : Prints information about the currently stored broadcast ephemeris data.

ssr : Prints information about the currently decoded PPP-B2b correction data.

stream : Displays the status of all active input/output data streams (e.g., connection status, byte counts).

shutdown: Terminates the rtppp process cleanly. Note: When data streams are configured using serial ports or TCP/IP connections, it is imperative to use this command to properly close connections and terminate the process, rather than abruptly closing the console window.

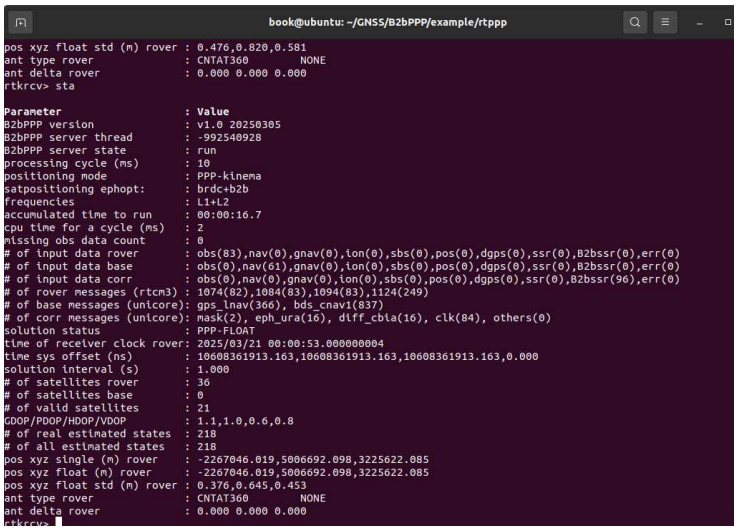
d. Example

Considering the nature of real-time data and the need for repeatable testing, an archived dataset specifically prepared for use with the RTKLIB-B2b software's playback mode is provided. (Details on RTKLIB's playback mode principles can be found in the RTKLIB documentation). This dataset is publicly available for download from the Zenodo archive at: https://doi.org/10.5281/zenodo.15105929

```
./rtppp -o ./conf/rtppp_25080_unicore.conf -s -t 0 -r 2
```

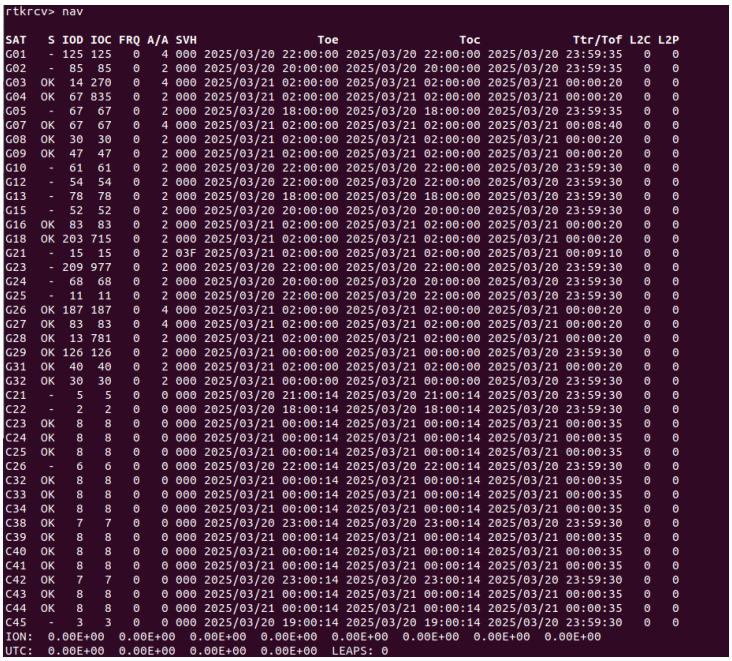
```
./rtppp -o ./conf/rtppp_25081_unicore.conf -s -t 0 -r 2
./rtppp -o ./conf/rtppp_25082_unicore.conf -s -t 0 -r 2
```

During console operation, users can monitor the solution status and related information by inputting commands at the prompt. The three figures below illustrate examples of the information output to the console upon executing the ‘sta’, ‘nav’, and ‘ssr’ commands.



```
book@ubuntu: ~/GNSS/B2bPPP/example/rtppp
pos xyz float std (m) rover : 0.476,0.820,0.581
ant type rover              : CNTAT360        NONE
ant delta rover             : 0.000 0.000 0.000
rtkrcv> sta

Parameter                   : Value
B2bPPP version              : v1.0 20250305
B2bPPP server thread        : -992540928
B2bPPP server state         : run
processing cycle (ms)       : 10
positioning mode            : PPP-kinema
satpositioning ephopt:      : brdc+b2b
frequencies                 : L1+L2
accumulated time to run     : 00:00:16.7
cpu time for a cycle (ms)   : 2
missing obs data count      : 0
# of input data rover       : obs(83),nav(0),gnav(0),ion(0),sbs(0),pos(0),dgps(0),ssr(0),B2bssr(0),err(0)
# of input data base        : obs(0),nav(61),gnav(0),ion(0),sbs(0),pos(0),dgps(0),ssr(0),B2bssr(0),err(0)
# of input data corr        : obs(0),nav(0),gnav(0),ion(0),sbs(0),pos(0),dgps(0),ssr(0),B2bssr(96),err(0)
# of rover messages (rtcm3) : 1074(82),1084(83),1094(83),1124(249)
# of base messages (unicore): gps_lnav(366), bds_cnav1(837)
# of corr messages (unicore): mask(2), eph_ura(16), diff_cbia(16), clk(84), others(0)
solution status             : PPP-FLOAT
time of receiver clock rover: 2025/03/21 00:00:53.000000004
time sys offset (ns)        : 10608361913.163,10608361913.163,10608361913.163,0.000
solution interval (s)       : 1.000
# of satellites rover       : 36
# of satellites base        : 0
# of valid satellites       : 21
GDOP/PDOP/HDOP/VDOP         : 1.1,1.0,0.6,0.8
# of real estimated states  : 218
# of all estimated states   : 218
pos xyz single (m) rover    : -2267046.019,5006692.098,3225622.085
pos xyz float (m) rover     : -2267046.019,5006692.098,3225622.085
pos xyz float std (m) rover : 0.376,0.645,0.453
ant type rover              : CNTAT360        NONE
ant delta rover             : 0.000 0.000 0.000
rtkrcv>
```

Fig 17. Example of the printout after running the 'sta' command in the RTKLIB-B2b real-time PPP (console mode).



```
rtkrcv> nav

SAT   S IOD IOC FRQ A/A SVH                 Toe                 Toc             Ttr/Tof L2C L2P
G01   - 125 125   0   4 000 2025/03/20 22:00:00 2025/03/20 22:00:00 2025/03/20 23:59:35   0   0
G02   -  85  85   0   2 000 2025/03/20 20:00:00 2025/03/20 20:00:00 2025/03/20 23:59:35   0   0
G03  OK  14 270   0   4 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G04  OK  67 835   0   2 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G05   -  67  67   0   2 000 2025/03/20 18:00:00 2025/03/20 18:00:00 2025/03/20 23:59:35   0   0
G07  OK  67  67   0   4 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:08:40   0   0
G08  OK  30  30   0   2 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G09  OK  47  47   0   2 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G10   -  61  61   0   2 000 2025/03/20 22:00:00 2025/03/20 22:00:00 2025/03/20 23:59:30   0   0
G12   -  54  54   0   2 000 2025/03/20 22:00:00 2025/03/20 22:00:00 2025/03/20 23:59:30   0   0
G13   -  78  78   0   2 000 2025/03/20 18:00:00 2025/03/20 18:00:00 2025/03/20 23:59:30   0   0
G15   -  52  52   0   2 000 2025/03/20 20:00:00 2025/03/20 20:00:00 2025/03/20 23:59:30   0   0
G16  OK  83  83   0   2 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G18  OK 203 715   0   2 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G21   -  15  15   0   2 03F 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:09:10   0   0
G23   - 209 977   0   2 000 2025/03/20 22:00:00 2025/03/20 22:00:00 2025/03/20 23:59:30   0   0
G24   -  68  68   0   2 000 2025/03/20 20:00:00 2025/03/20 20:00:00 2025/03/20 23:59:30   0   0
G25   -  11  11   0   2 000 2025/03/20 22:00:00 2025/03/20 22:00:00 2025/03/20 23:59:30   0   0
G26  OK 187 187   0   4 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G27  OK  83  83   0   4 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G28  OK  13 781   0   2 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G29  OK 126 126   0   2 000 2025/03/21 00:00:00 2025/03/21 00:00:00 2025/03/20 23:59:30   0   0
G31  OK  40  40   0   2 000 2025/03/21 02:00:00 2025/03/21 02:00:00 2025/03/21 00:00:20   0   0
G32  OK  30  30   0   2 000 2025/03/21 00:00:00 2025/03/21 00:00:00 2025/03/20 23:59:30   0   0
C21   -   5   5   0   0 000 2025/03/20 21:00:14 2025/03/20 21:00:14 2025/03/20 23:59:30   0   0
C22   -   2   2   0   0 000 2025/03/20 18:00:14 2025/03/20 18:00:14 2025/03/20 23:59:30   0   0
C23  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C24  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C25  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C26   -   6   6   0   0 000 2025/03/20 22:00:14 2025/03/20 22:00:14 2025/03/20 23:59:30   0   0
C32  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C33  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C34  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C38  OK   7   7   0   0 000 2025/03/20 23:00:14 2025/03/20 23:00:14 2025/03/20 23:59:30   0   0
C39  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C40  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C41  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C42  OK   7   7   0   0 000 2025/03/20 23:00:14 2025/03/20 23:00:14 2025/03/20 23:59:30   0   0
C43  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C44  OK   8   8   0   0 000 2025/03/21 00:00:14 2025/03/21 00:00:14 2025/03/21 00:00:35   0   0
C45   -   3   3   0   0 000 2025/03/20 19:00:14 2025/03/20 19:00:14 2025/03/20 23:59:30   0   0
ION:  0.00E+00  0.00E+00  0.00E+00  0.00E+00  0.00E+00  0.00E+00  0.00E+00  0.00E+00
UTC:  0.00E+00  0.00E+00  0.00E+00  0.00E+00  LEAPS: 0
```

Fig 18. Example of the printout after running the 'nav' command in the RTKLIB-B2b real-time PPP (console mode).

```
solution interval (s)        : 1.000
# of satellites rover        : 37
# of satellites base         : 0
# of valid satellites        : 19
GDOP/PDOP/HDOP/VDOP          : 1.3,1.1,0.6,1.0
# of real estimated states   : 218
# of all estimated states    : 218
pos xyz single (m) rover     : -2267045.249,5006691.596,3225621.816
pos xyz float (m) rover      : -2267045.249,5006691.596,3225621.816
pos xyz float std (m) rover  : 0.030,0.036,0.022
ant type rover               : CNTAT360        NONE
ant delta rover              : 0.000 0.000 0.000
rtkrcv> ssr

SAT   S UDI IODSSR URA                  T0   D0-A   D0-C   D0-R       C0     C1     C2
G03  OK  48   1     39 2025/03/21 00:13:41  0.019 -0.141 -0.102   -2.987  0.000 0.0000
G04  OK  48   1     39 2025/03/21 00:13:41  0.144 -1.011 -0.365    0.125  0.000 0.0000
G07  OK  48   1     39 2025/03/21 00:13:41  0.282  0.038  0.525    0.000  0.000 0.0000
G08  OK  48   1     39 2025/03/21 00:13:41  0.064 -0.410  0.045   -0.109  0.000 0.0000
G09  OK  48   1     39 2025/03/21 00:13:41  0.050  0.006  0.448   -0.658  0.000 0.0000
G16  OK  48   1     39 2025/03/21 00:13:41 -0.029  1.126  0.102    1.331  0.000 0.0000
G18  OK  48   1     37 2025/03/21 00:13:41 -0.126 -0.416  1.133   -1.230  0.000 0.0000
G26  OK  48   1     39 2025/03/21 00:13:41 -0.237  1.274  0.371    0.477  0.000 0.0000
G27  OK  48   1     31 2025/03/21 00:13:41 -0.250  0.154 -0.147   -0.763  0.000 0.0000
G28  OK  48   1     39 2025/03/21 00:13:41 -0.174 -0.326 -0.019    0.491  0.000 0.0000
G31  OK  48   1     39 2025/03/21 00:13:41  0.093 -1.747 -0.832    0.078  0.000 0.0000
C23  OK  48   1     31 2025/03/21 00:13:41  0.005 -0.077  0.090   -0.274  0.000 0.0000
C24  OK  48   1     31 2025/03/21 00:13:41 -0.046 -0.058 -0.090   -0.002  0.000 0.0000
C25  OK  48   1     31 2025/03/21 00:13:41 -0.008 -0.134 -0.026   -0.552  0.000 0.0000
C32  OK  48   1     31 2025/03/21 00:13:41  0.011 -0.077 -0.122    0.470  0.000 0.0000
C33  OK  48   1     31 2025/03/21 00:13:41 -0.010 -0.051 -0.122    0.168  0.000 0.0000
C34  OK  48   1     31 2025/03/21 00:13:41 -0.037  0.026  0.083   -0.443  0.000 0.0000
C39  OK  48   1     31 2025/03/21 00:13:41 -0.006  0.000 -0.160   -0.616  0.000 0.0000
C40  OK  48   1     23 2025/03/21 00:13:41 -0.002  0.102  0.147    0.467  0.000 0.0000
C41  OK  48   1     31 2025/03/21 00:13:41 -0.011  0.038 -0.128   -0.274  0.000 0.0000
C43  OK  48   1     31 2025/03/21 00:13:41 -0.016  0.000  0.128    0.306  0.000 0.0000
C44  OK  48   1     31 2025/03/21 00:13:41 -0.046 -0.019  0.058   -0.402  0.000 0.0000
rtkrcv>
```

Fig 19. Example of the printout after running the 'ssr' command in the RTKLIB-B2b real-time PPP (console mode).

The resulting positioning solutions stored in the .pos file can be directly plotted and analyzed using RTKLIB's rtkplot tool. For validation purposes, reference coordinates were established using final precise orbit and clock products published by the European Space Agency (ESA). These coordinates represent the average values obtained from converged static PPP solutions. In the Earth-Centered Earth-Fixed (ECEF) frame, these reference coordinates are : X = -2267045.334 m Y = 5006691.635 m Z = 3225621.891 m

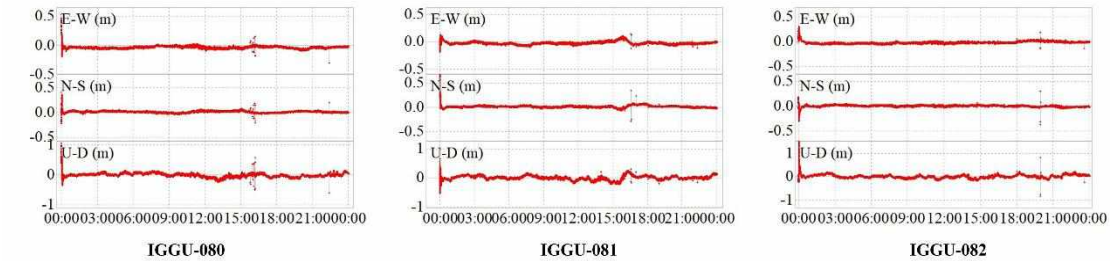


Fig 20. Example results Plotted using RTKLIB's rtkplot.

5. Software Maintenance and Data Sharing

(1) Software Maintenance

With the ongoing development of the BeiDou Navigation Satellite System, PPP-B2b services are expected to support more satellite systems and correction types in the future. We will continue to maintain this software on GitHub. We sincerely welcome users to report any issues encountered during usage by creating an issue. To facilitate a quicker resolution, we recommend using the following template:

```
## Issue Type
[ ] Data Parsing Anomaly
[ ] Positioning Result Deviation
[ ] Other Issues

## Conditions to Reproduce
1. Receiver Model: _____
2. Data File: _____ (If using an example provided with the software, just provide the filename. For other data, please provide a secure download link)
3. Operating Environment: _____

## Key Log Information
(Paste/upload snippets of log messages or errors you believe are relevant. If using an IDE, we currently only accept error reports related to VS Code.)

## Expected Behavior
(Clearly describe the expected outcome or result)

## Additional Information
(Screenshots, etc. Format is flexible)
```

(2) Real-time PPP-B2b and Navigation Data Stream Sharing Service

Considering that some users may not have convenient long-term access to a PPP-B2b receiver for data collection, we provide a sharing service for real-time PPP-B2b and ephemeris data streams. Our antenna is located on the rooftop of a building within the campus of the Innovation Academy for Precision Measurement Science and Technology (APM) in Wuhan, China. Data is received using a SINO_K803W receiver and a CNT AT360 antenna, and broadcast externally via an NtripCaster.

The free test account details are provided below. Should there be any changes, we will update this section of the documentation on GitHub.

```
Address：1.94.195.74
Port：2101
Test Account Username：user2
Test Account Password：pass2
PPP-B2b Information Mount Point：SINO_B2b
Ephemeris Information Mount Point: SINO_Nav
```

Please note: This account is provided solely for the purpose of testing PPP-B2b and ephemeris information decoding. It must not be

used for any other purpose. While we will strive to maintain its stability, we reserve the right to deactivate this account at any time without prior notice to users.

(3) GNSS Datasets for RTKLIB-B2b & RTKLIB Playback and Post-Processing

This public archived dataset collection contains two complementary GNSS datasets designed to support research, algorithm development, and software testing/validation (e.g., for RTKLIB-B2b, RTKLIB) involving Precise Point Positioning (PPP) utilizing BeiDou PPP-B2b corrections. These two datasets cater specifically to two typical application scenarios: real-time playback simulation and post-processing analysis.

a. Dataset 1: SINO & UNICORE Receiver Raw Log Data for Real-time Playback

To provide binary format data, as output directly by receivers, required for simulating real-time data streams; suitable for testing the real-time processing (playback functionality) of software such as RTKLIB-B2b and RTKLIB.

Data was collected on the rooftop of Building 1 at the Innovation Academy for Precision Measurement Science and Technology (APM), CAS, Wuhan, China, using two types of GNSS receivers, SinoGNSS (SINO) K803W and Unicore Communications (UNICORE) UM980, along with a CNT AT360 antenna.

Time Period Covered: Day of Year (DOY) 080 to 082, 2025 (corresponding to March 21-23, 2025).

It is now available for users to download in Zenodo:

https://doi.org/10.5281/zenodo.15105929

b. Dataset 2: High-Rate MGEX Observations, DLR Ephemeris, and Unicore B2b Correction Data for Post-Processing

To provide data in standard or common formats suitable for PPP post-processing analysis, with a particular focus on evaluating the performance and application effectiveness of PPP-B2b correction services.

Observation data originates from three IGS MGEX stations: gamg, ulab, and wuh2. Broadcast ephemeris is sourced from the German Aerospace Center (DLR). PPP-B2b correction data comes from independent logging by a Unicore receiver. Observation data covers DOY 065-067, 2025. Broadcast ephemeris and PPP-B2b correction data

cover DOY 065-068, 2025 (corresponding to March 6-9, 2025).

It is now available for users to download in Zenodo:

https://doi.org/10.5281/zenodo.15110885

6. References

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7. Appendix

(1) PPP-B2b Format Description

The standard format specification for PPP-B2b messages has been detailed comprehensively in the official documentation. In this section, we focus solely on describing the message organization employed in the custom PPP-B2b formats implemented by two specific vendors: Sino and Unicore.

a. SINO

The complete data manual is shown in detail:

https://www.sinognss.com/service/download/

The PPP-B2b message structure utilized by Sino receivers is fundamentally based on the standard format, but introduces modifications primarily within the message header information. A diagram illustrating the correspondence between the standard format organization and the Sino organization is provided below.

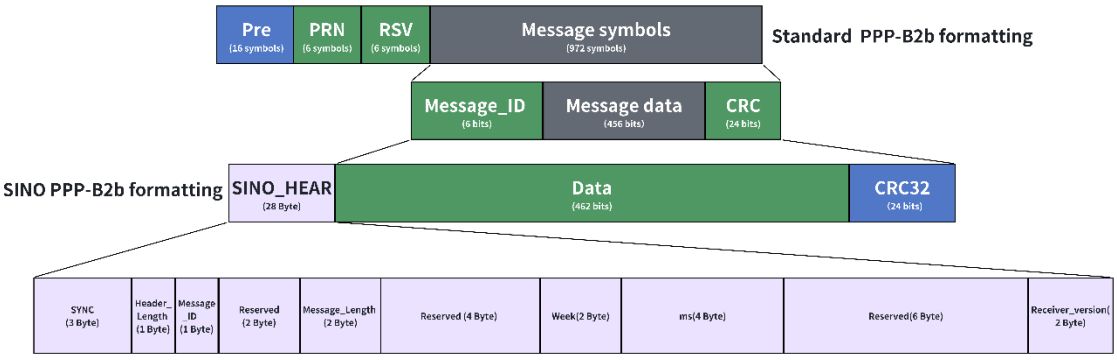

Fig Comparison of kinematic PPP convergence times across different stations using various satellite system combinations, where 'C' denotes BDS and 'G' denotes GPS.

b. UNICORE

The complete data manual is shown in detail:

https://www.unicorecomm.com/download

The organization of PPP-B2b messages from Unicore receivers differs more significantly from the standard format. Generally, in the standard specification, the sequence composed of the Preamble, PRN (Pseudo-Random Noise code identifying the satellite transmission source), and Reserved bits (Pre+PRN+RSV) is considered the message header, while the subsequent "Message Symbols" constitute the message data payload. In the Unicore PPP-B2b implementation, the PRN number (identifying the

transmitting satellite) is relocated from the header section into the message data payload. Conversely, the Message Type identifier (Message ID) is stored within the header section. The relationship between these two structures is depicted in the figure below. However, it is important to note that even within the message data payload section, certain differences in the sequential order of specific data fields may still exist compared to the standard. Users should consult the official Unicore message format documentation for precise details.

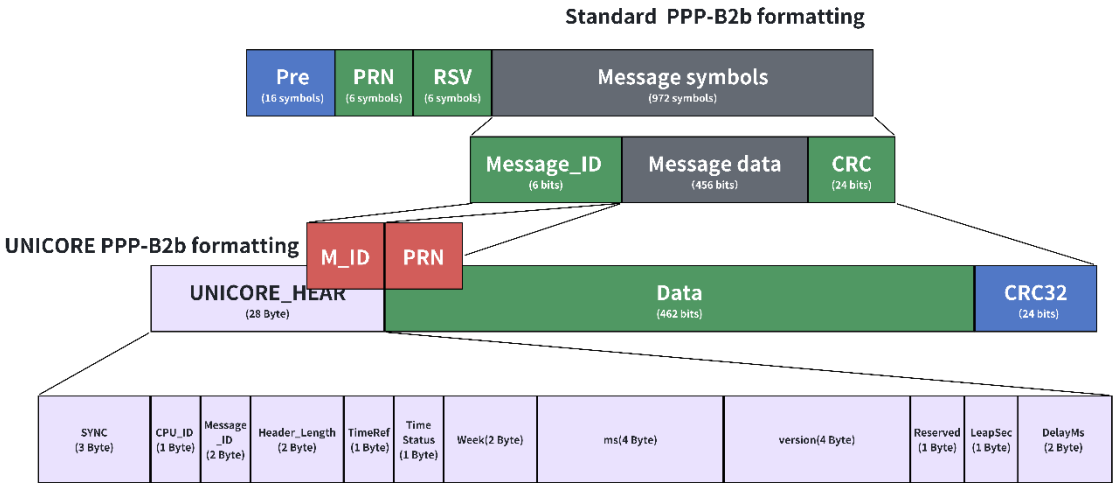

Fig Comparison of kinematic PPP convergence times across different stations using various satellite system combinations, where 'C' denotes BDS and 'G' denotes GPS.

(2) Reference Commands for Receiver Observation, Ephemeris, and PPP-B2b Data Output

This section provides reference command sequences for configuring SinoGNSS and Unicore Communications receivers to output the necessary data streams for observation processing, ephemeris collection, and PPP-B2b corrections via specified communication ports (COM1, COM2, COM3).

a. SINO

The following sequence configures a Sino receiver:

```
# Clear existing logs on COM ports
Unlogall COM1
Unlogall COM2
Unlogall COM3

# Log RTCM SSR and Observation messages on COM1
Log com1 rtcm1074b ontime 5
Log com1 rtcm1084b ontime 5
```

```
Log com1 rtcm1094b ontime 5
Log com1 rtcm1124b ontime 5
# Log RTCM Ephemeris messages on COM1 (adjust interval as needed)
Log com1 rtcm1019b ontime 90
Log com1 rtcm1020b ontime 90
Log com1 rtcm1045b ontime 90
Log com1 rtcm1042b ontime 90

# Log Proprietary Ephemeris messages on COM2 (alternative/redundant)
Log com2 bd3ephemb ontime 1
Log com2 gpsephemb ontime 1

# Log Raw PPP-B2b Subframe data on COM3
Log com3 B2BRAWNAVSUBFRAMEb ontime 1

# Save the configuration
Saveconfig
```

b. UNICORE

The following sequence configures a Unicore Communications receiver:

```
# Clear existing logs on COM ports
unlog com1
unlog com2
unlog com3

# Log RTCM SSR messages on COM1 (Interval 1 second)
rtcm1074 com1 1
rtcm1084 com1 1
rtcm1094 com1 1
rtcm1124 com1 1

 # Enable B2b-PPP mode
CONFIG PPP ENABLE B2b-PPP

# Log Proprietary PPP-B2b messages on COM2 (Output on change)
PPPB2BINFO1B COM2 ONCHANGED
PPPB2BINFO2B COM2 ONCHANGED
PPPB2BINFO3B COM2 ONCHANGED
PPPB2BINFO4B COM2 ONCHANGED

# Log Proprietary Ephemeris messages on COM3 (Interval 5 seconds)
GPSEPHB COM3 5
BD3EPHB COM3 5

# Save the configuration
```

```
saveconfig
```